

Pave The Way

Road Mapping Using Machine Learning



PAVE THE WAY
PTW.ml

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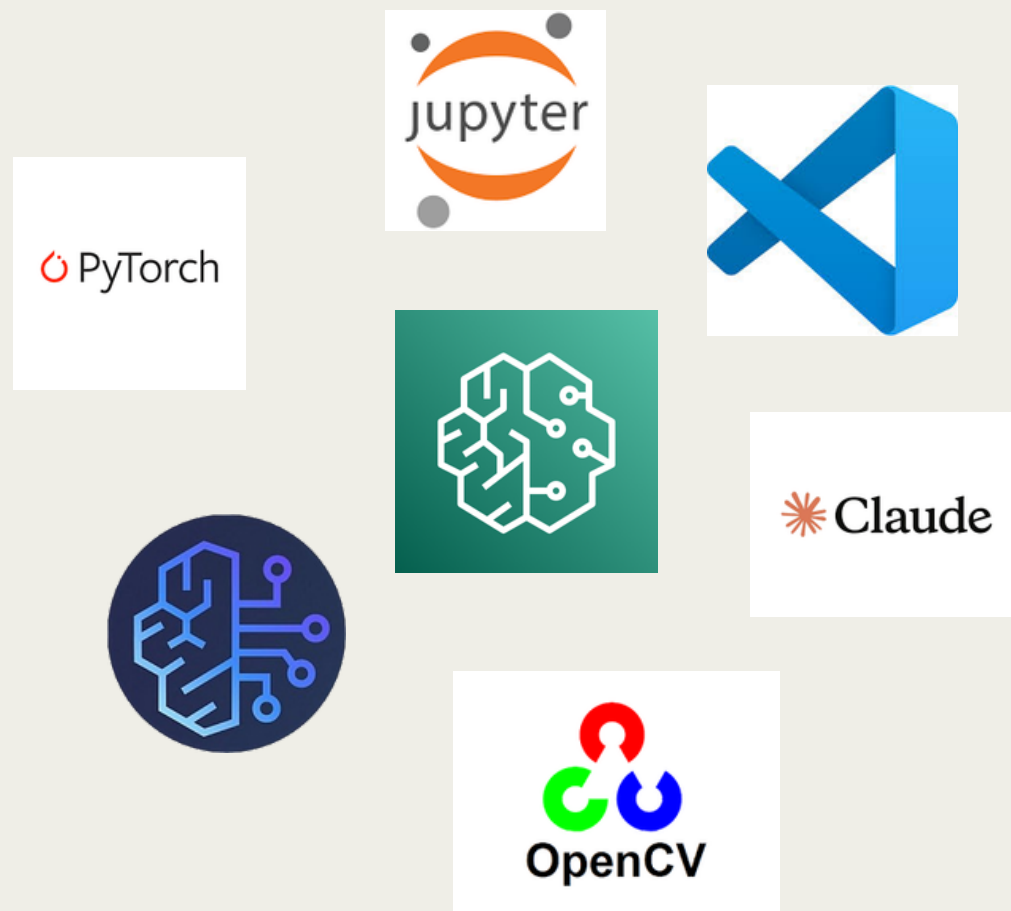
Nhat Chu - Graduate Student @ CSU

Brice Davis - Senior @ CSU

USDA x NASA
Hackathon
2025



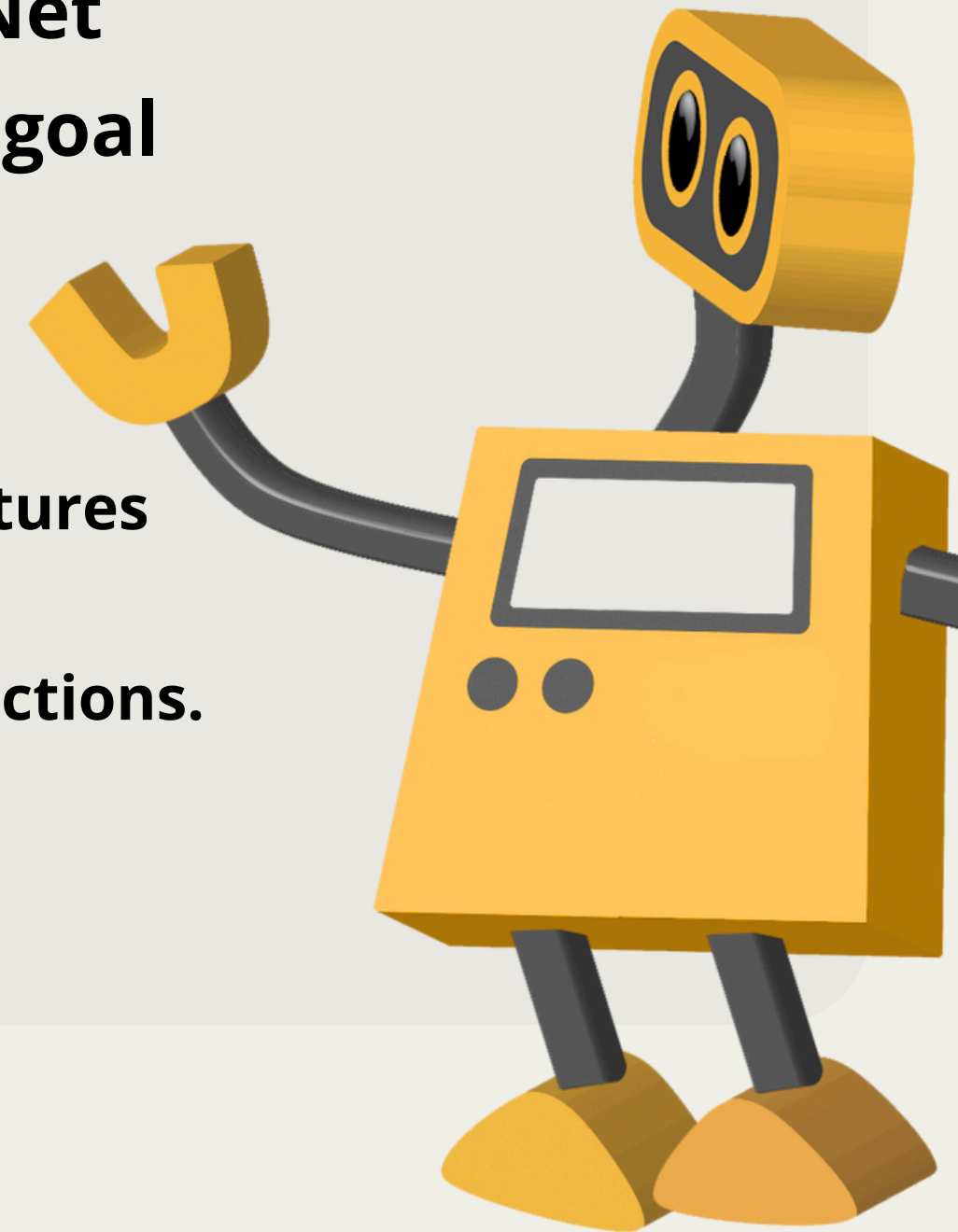
What We Used

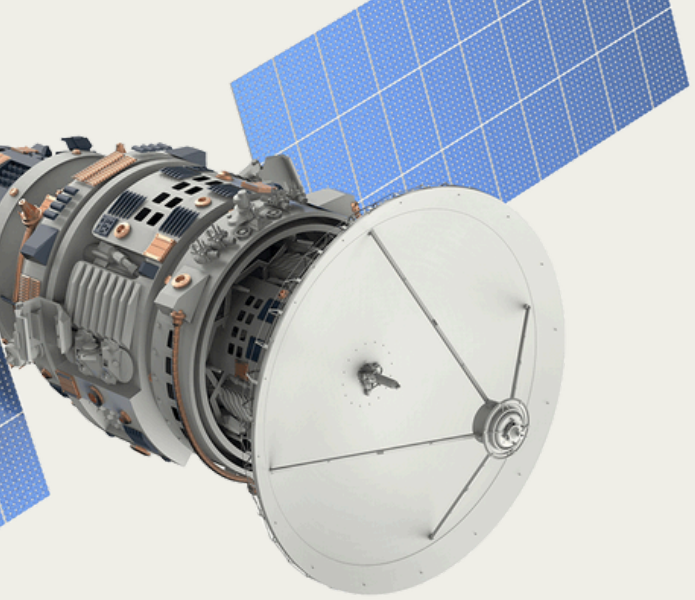


- 1 PyTorch, Cv2
- 2 AWS SageMaker
- 3 LLM'S e.g BedRock, Claude
- 4 VScode, Jyp
- 5 Cuda, ml.g5.xlarge

Why UNET?

- 1** We're trying to label each pixel as road or non-road. U-Net is specifically designed for dense prediction, where the goal is to assign a label to every pixel in an image
- 2** Encoder: Captures context and high-level features (e.g. where structures like roads might be).
Decoder: Restores spatial resolution and generates pixel-level predictions.





What does PTW do?

01

**Pre-
Processing**

02

**Data
splitting/se
gmentation**

03

**Model
predicts
and
arranges a
binary map**

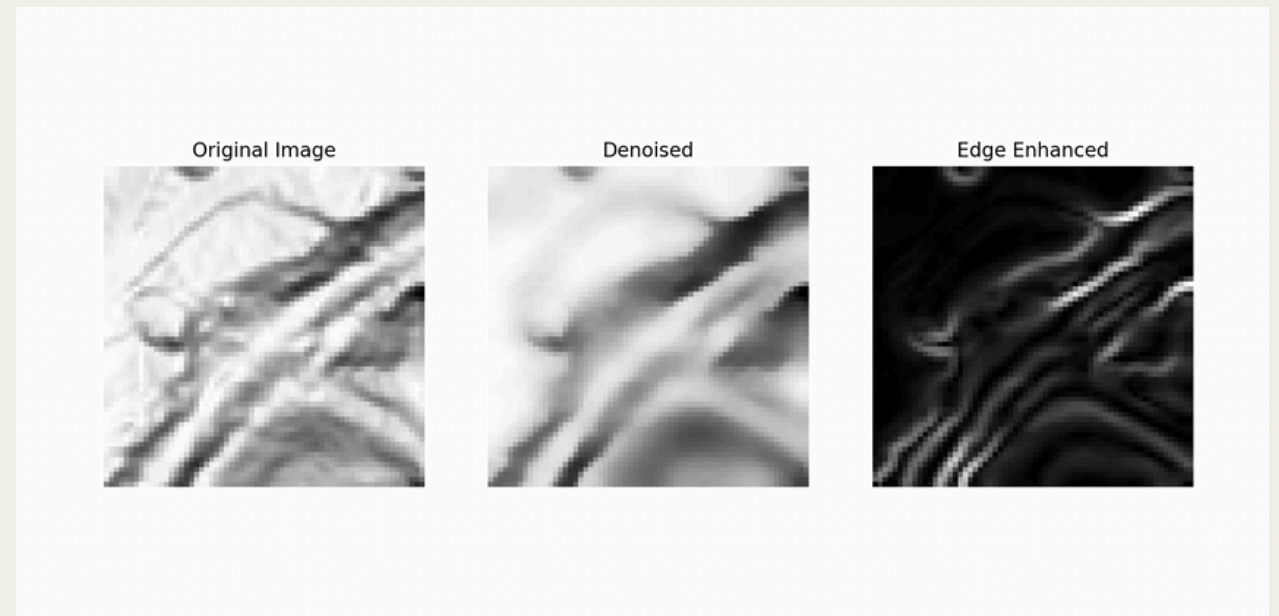
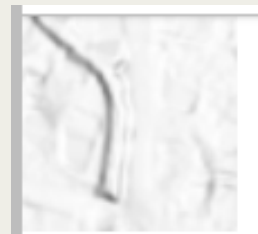
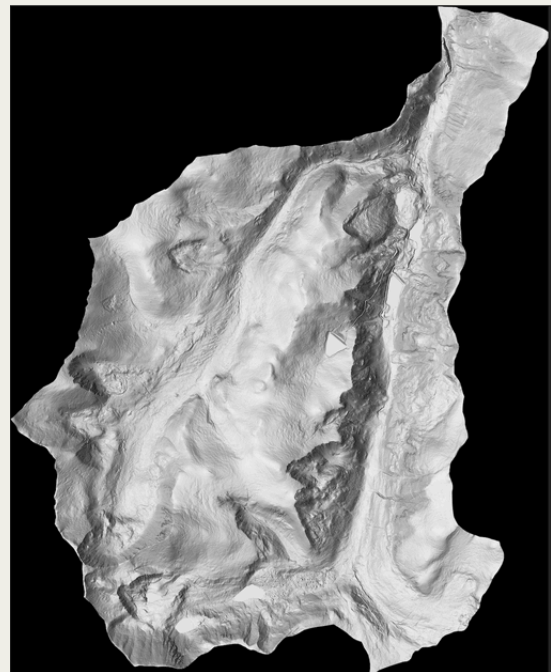
04

**Overlays
the binary
map with
original
image**



Preprocessing and data splitting

- 1 Data segmentation/Splitting
- 2 Image Edge Enhancement
- 3 Packaging the data for training



Original Image

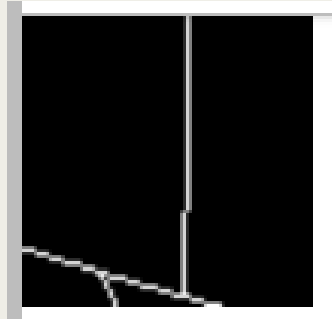
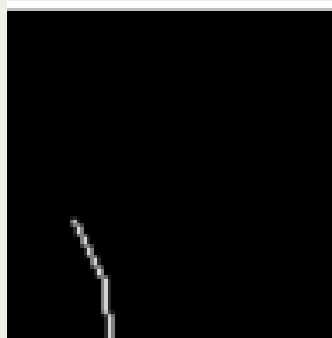
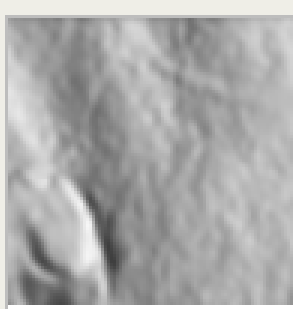
Image Patches

Noise Reduction and Edge Enhancing

Binary Mapping

00

experimenting with patch sizes,
(512,256,128,64) and stride sizes



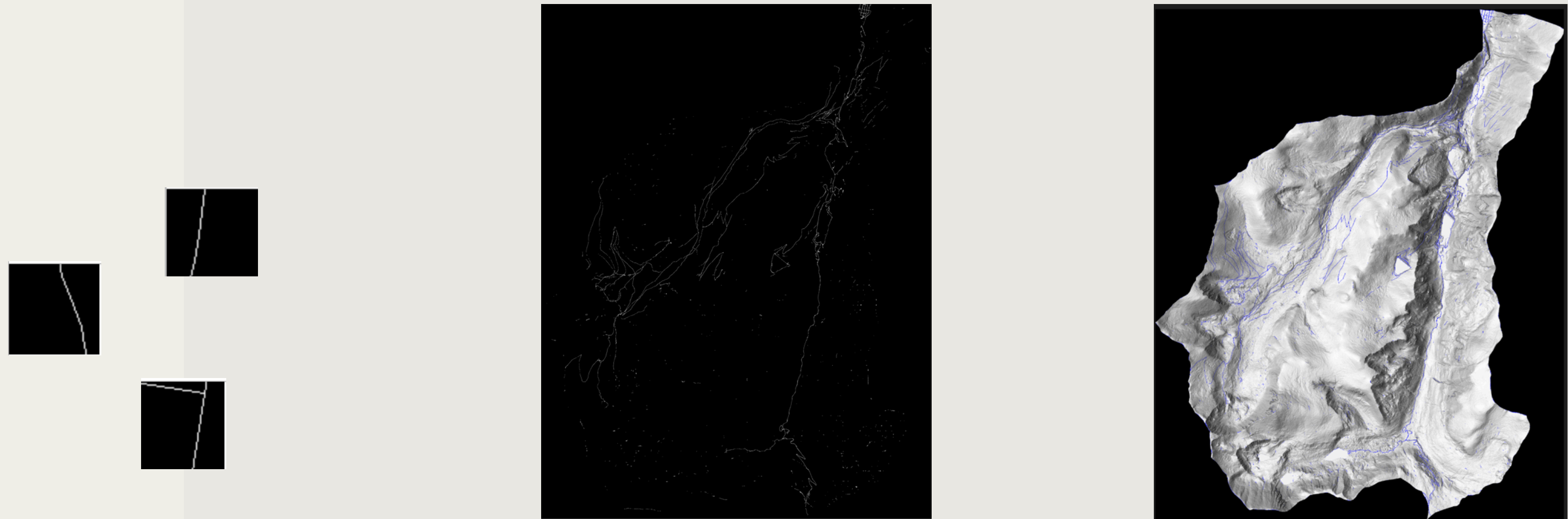
01

Our model can predict a binary maps
based on the inputted data

10

Adjusting thresholds to reduce
noise in post

Patch Realignment And Overlay



Predicted Binary Patches

Assembled Binary Map

Binary Map Overlay

Costs and Time

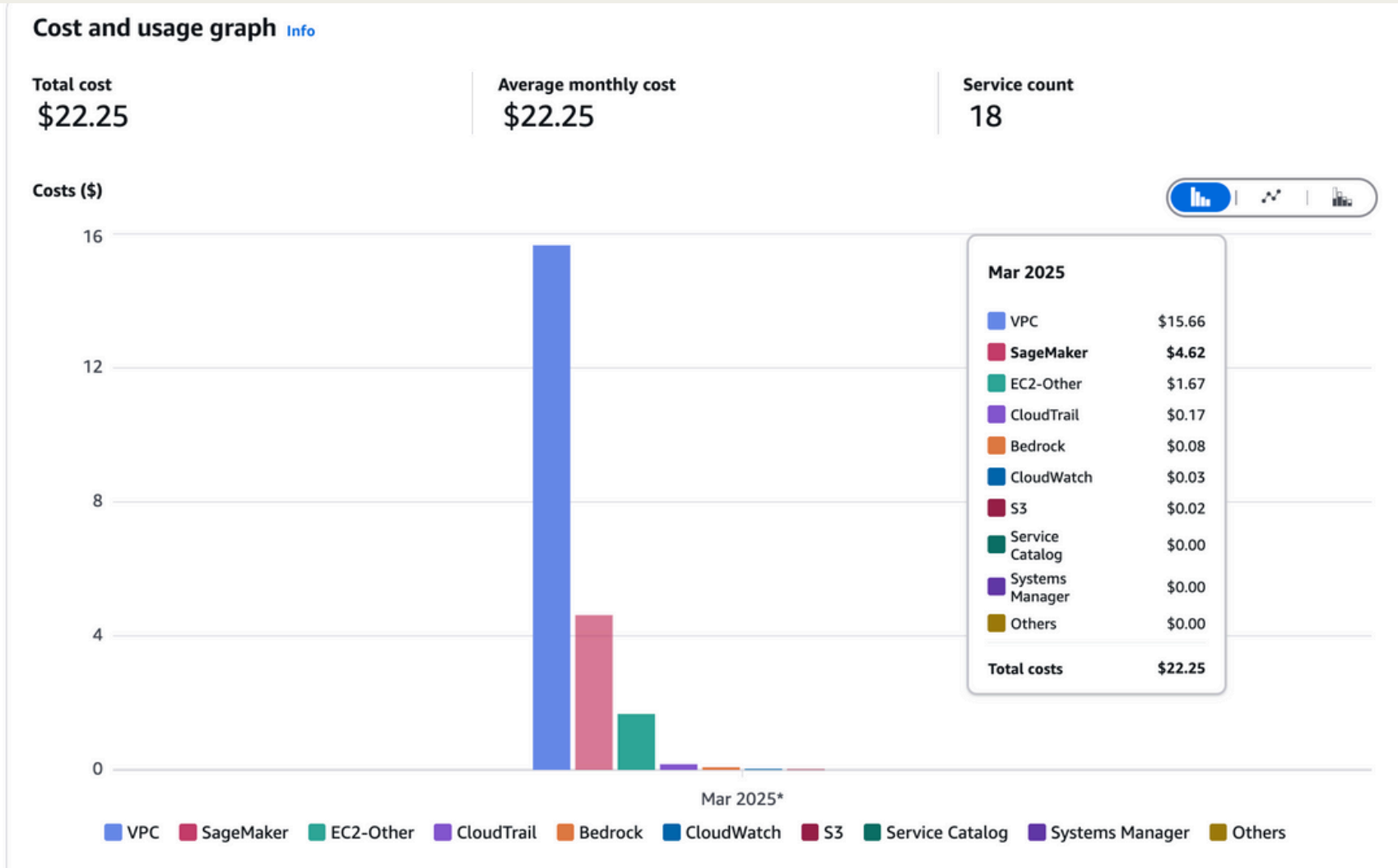
\$1.23 per train

Current month costs

\$22.25

Forecasted month end costs

\$16.01



Service	Service total	March 2025*
Total costs	\$22.25	\$22.25
VPC	\$15.66	\$15.66
SageMaker	\$4.62	\$4.62
EC2-Other	\$1.67	\$1.67
CloudTrail	\$0.17	\$0.17
Bedrock	\$0.08	\$0.08
CloudWatch	\$0.03	\$0.03
S3	\$0.02	\$0.02
Service Catalog	\$0.00	\$0.00
Systems Manager	\$0.00	\$0.00
Key Management Service	\$0.00	\$0.00
Lambda	\$0.00	\$0.00
SQS	\$0.00	\$0.00
Elastic File System	\$0.00	\$0.00
Step Functions	\$0.00	\$0.00
CloudFormation	\$0.00	\$0.00
Glue	\$0.00	\$0.00
EC2-Instances	\$0.00	\$0.00
SNS	\$0.00	\$0.00

Questions?